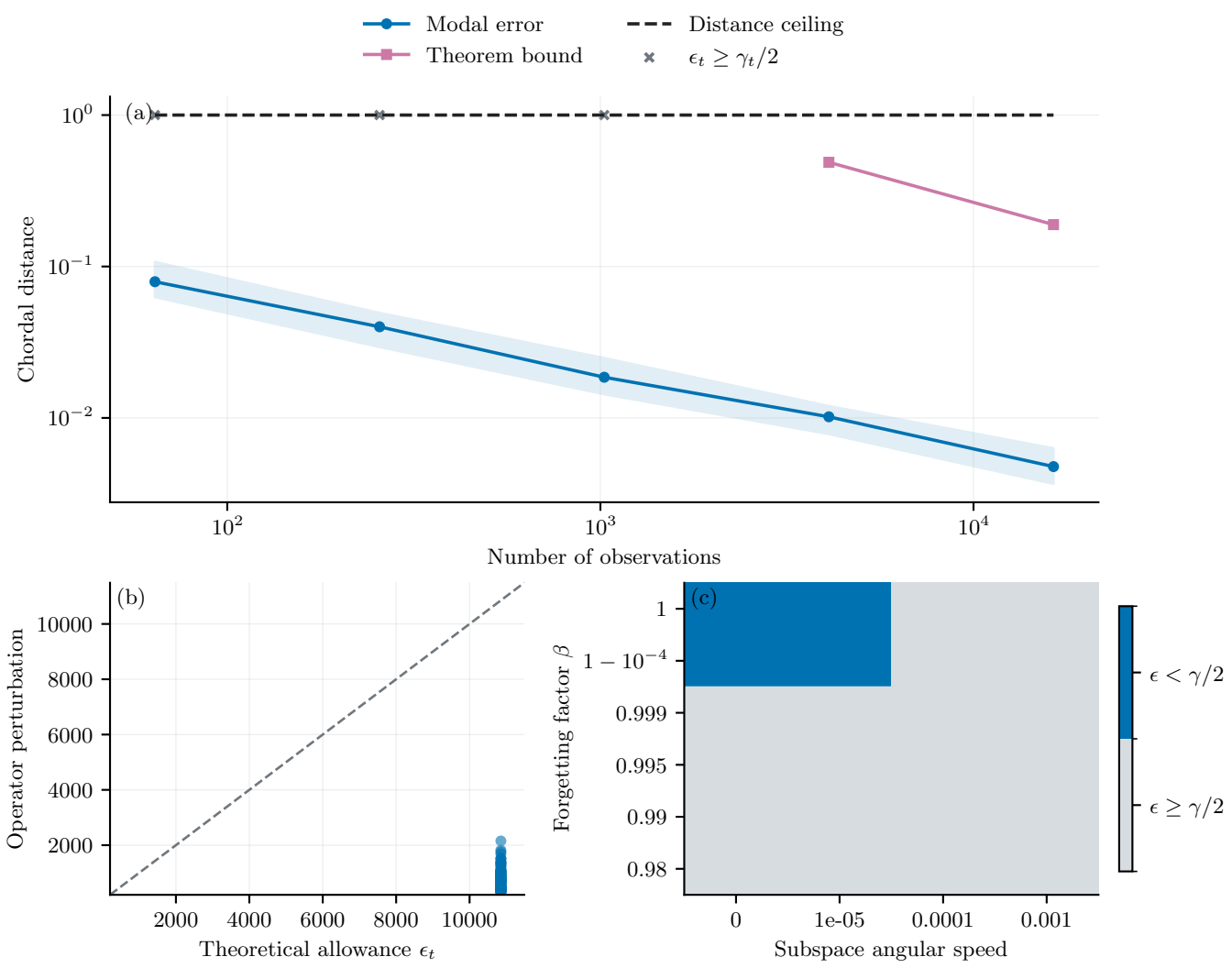
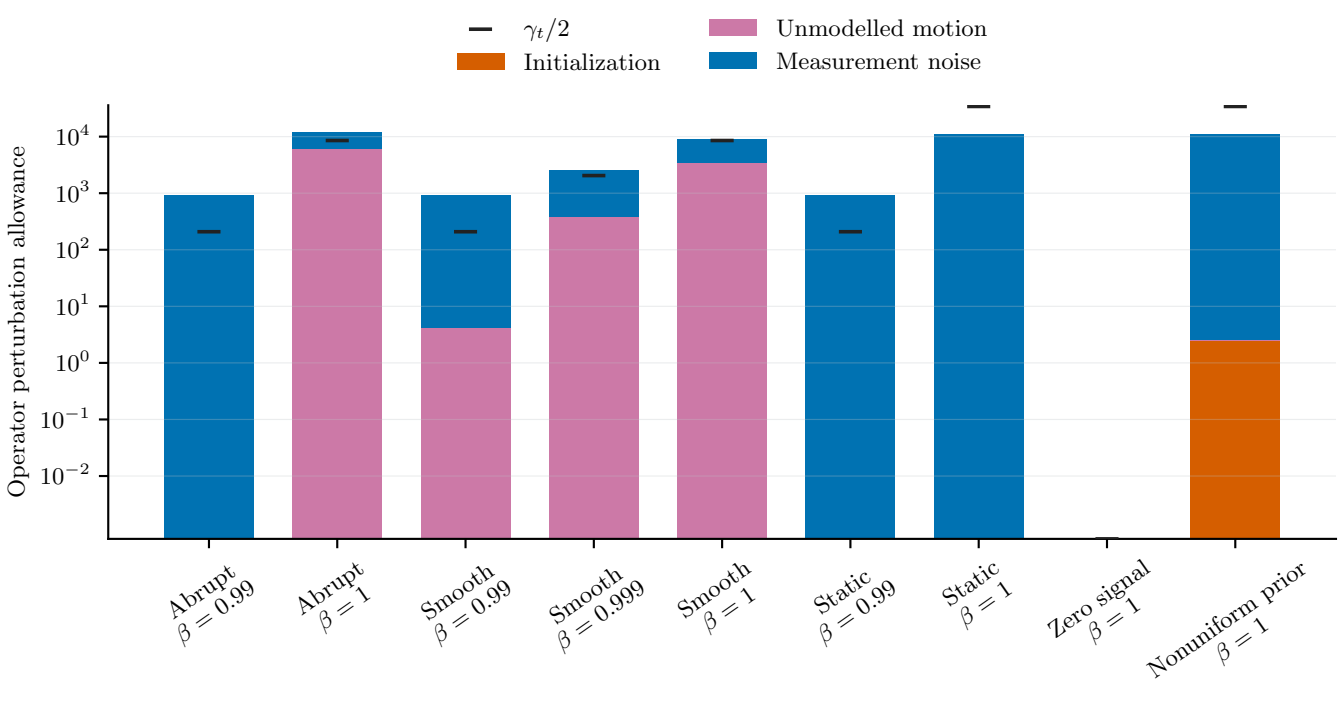
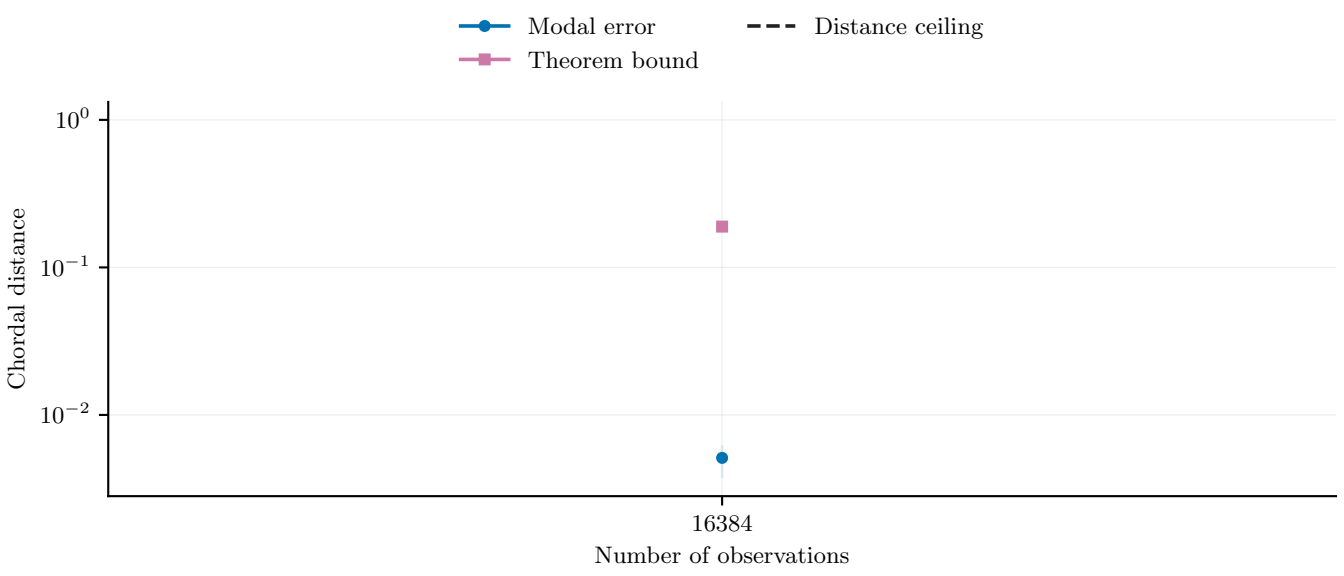
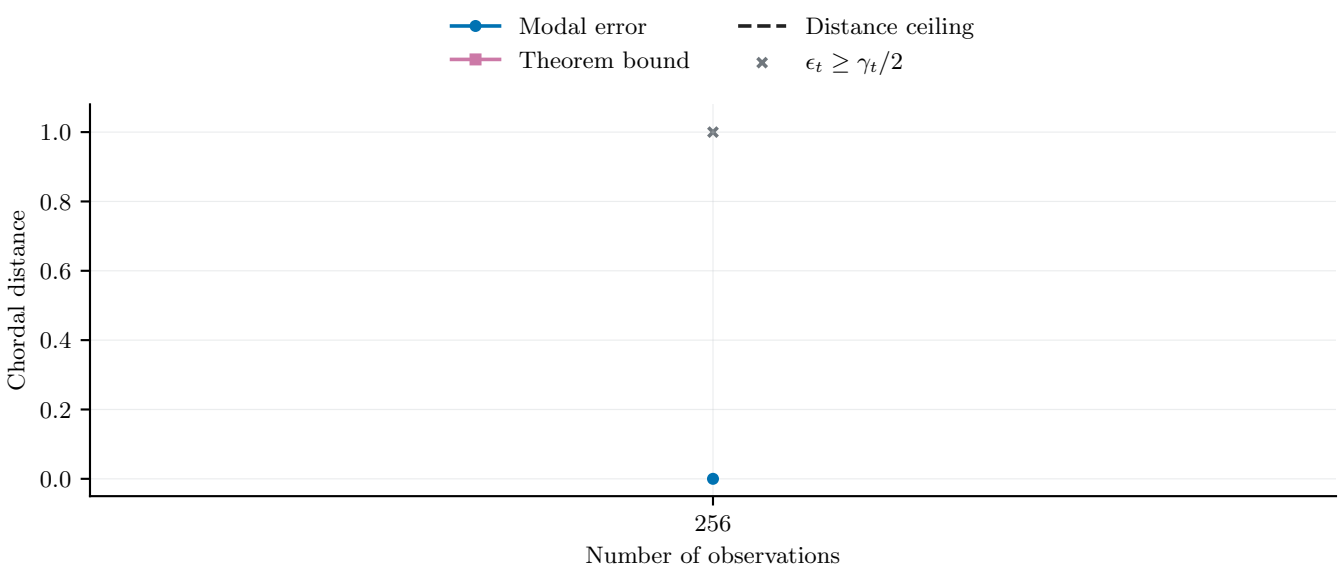

Modal error
Theorem bound
Distance ceiling
$\epsilon_t \geq \gamma_t/2$
(a)
Chordal distance
10^0
10^{-1}
10^{-2}
10^2
10^3
10^4
Number of observations
(b)
Operator perturbation
10000
8000
6000
4000
2000
2000
4000
6000
8000
10000
Theoretical allowance ϵ_t
(c)
Forgetting factor β
1
$1 - 10^{-4}$
0.999
0.995
0.99
0.98
0
1e-05
0.0001
0.001
Subspace angular speed
$\epsilon < \gamma/2$
$\epsilon \geq \gamma/2$


$\gamma_t/2$
Initialization
Unmodelled motion
Measurement noise
Operator perturbation allowance
10^4
10^3
10^2
10^1
10^0
10^{-1}
10^{-2}
Abrupt $\beta = 0.99$
Abrupt $\beta = 1$
Smooth $\beta = 0.99$
Smooth $\beta = 0.999$
Smooth $\beta = 1$
Static $\beta = 0.99$
Static $\beta = 1$
Zero signal $\beta = 1$
Nonuniform prior $\beta = 1$


Modal error
Distance ceiling
Theorem bound
10^{0}
10^{-1}
10^{-2}
Chordal distance
16384
Number of observations


Modal error
Theorem bound
Distance ceiling
$\epsilon_t \geq \gamma_t/2$
Chordal distance
1.0
0.8
0.6
0.4
0.2
0.0
256
Number of observations